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To the Editor, *Scientific Reports*

Dear Editor,

Please find our revised manuscript, “BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Biological Prior Knowledge”, for reconsideration in *Scientific Reports*. We thank the Editor and Reviewers for their detailed assessments. The revision is accompanied by a clean manuscript, a marked manuscript, Supporting Information, and a point-by-point response.

BioInteract addresses drug–target interaction prioritisation with a residue-level cross-attention model while explicitly separating model-native attribution from structural evidence. The revised manuscript is suited to *Scientific Reports* because it provides reproducible computational methods, released code and artifacts, diagnostic analyses of dataset scope, and a claim-calibrated evaluation of an AI-enabled drug-discovery workflow.

The substantive revisions include an audit of the original identifier-based splits, exact-sequence-grouped within-Davis evaluations, and a pre-specified strict BindingDB external evaluation. We restored the Davis baseline comparison and component-removal results, regenerated the associated figures from the reported values, and reconciled Figure 1 Panel C with Table 1 by including all seven comparison methods. We also removed the unsupported nominal ABL1 variant-comparison and hotspot-count analyses, tightened the interpretation of attention and Grad-CAM outputs, and expanded the documentation of code, configuration, and evaluation provenance.

For transparency, several requests that would require a new independently benchmarked study are not represented as completed experiments. Specifically, we report a full-length identity audit and stratified cold-target results but do not claim a new 60%-clustered split; we define functional-group attribution but do not claim a matched-analogue stability test; we cite recent methods without claiming unmatched reimplementations; we state the implementation environment without presenting unmeasured BindingDB-scale profiling; and we do not claim a PDDBind/LP-PDDBind overlap or structural-contact validation. In each case, the manuscript and response letter state the scope boundary and avoid conclusions that would require the absent evidence.

Thank you for considering our revision.

Sincerely,
Zhi Huang, on behalf of all authors